Software Requirements Specification Document (SRS)

(Aligned with ISO/IEC/IEEE 29148:2018)

Document Organization and Conventions

This Software Requirements Specification (SRS) document is structured in alignment with **ISO/IEC/IEEE 29148:2018 – Systems and Software Engineering — Life Cycle Processes — Requirements Engineering**. The document defines functional and non-functional requirements in a clear, consistent, and testable manner. Each section addresses a distinct aspect of the Customatch system. Requirements are stated using "shall" statements where applicable to ensure clarity and verifiability. Numbering is used for traceability.

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1. Introduction

1.1 Purpose

The purpose of this document is to formally specify the software requirements for **Customatch**, an interactive online customization and provider-matching platform. This SRS is intended to serve as a contractual, technical, and managerial reference for all stakeholders involved in the system's lifecycle, including developers, designers, testers, business stakeholders, and potential investors.

1.2 Scope

Customatch enables customers to visually customize products or services and connect with qualified providers capable of delivering the customized outcome. The initial release focuses on **cake customization** as the Minimum Viable Product (MVP). Users select cake attributes such as shape, size, flavor, icing, and decoration, visualize the result, and place orders with verified bakers. The architecture supports expansion into additional personalization domains, including furniture, fashion, automotive interiors, and event design.

1.3 Product OverviewCustomatch is a B2B2C digital marketplace combining:

- Visual customization tools (engine)
- Pricing rules engine
- Intelligent marketplace matching system
- Order and transaction handling
- A communication layer
- Market intelligence and analytics

The platform acts as an intermediary between customers and providers while structuring creativity into executable, price-aware designs.

1.4 Intended Audience

This SRS is intended for the following audiences:

- **Software Developers and Engineers**: To understand detailed functional and non-functional requirements for implementing the Customatch platform according to agreed specifications.
- **UI/UX Designers**: To design intuitive, user-centered interfaces that align with the customization workflows and usability requirements defined in this document.
- **Quality Assurance and Testers**: To derive test cases, validation criteria, and acceptance tests based on clearly defined and measurable requirements.
- **Product Managers and Project Managers**: To plan development milestones, manage scope, track progress, and ensure alignment between business goals and technical execution.
- **Business Stakeholders and Founders**: To validate that the software requirements support the Customatch business model, MVP goals, and long-term vision.
- **Investors and Partners**: To gain a high-level yet structured understanding of the system's capabilities, scalability, and feasibility.
- **Regulatory and Compliance Reviewers**: To assess whether the platform aligns with applicable e-commerce, data protection, and consumer protection requirements.
- **End Users (Customers, Providers)**: To understand system expectations and ensure the platform meets real-world customization, ordering, and fulfillment needs.

This document assumes the audience has a basic understanding of web-based systems, but does not require deep technical expertise unless otherwise specified.

1.5 Definitions, Acronyms, and Abbreviations

- **SRS**: Software Requirements Specification — a formal document describing system requirements
- **MVP**: Minimum Viable Product — the initial version of the system with core features
- **PWA**: Progressive Web Application — a web application with app-like capabilities such as offline access and installation
- **UAT**: User Acceptance Testing — validation performed by end users to confirm the system meets requirements
- **UI**: User Interface — visual components used to interact with the system
- **UX**: User Experience — overall ease of use and satisfaction when interacting with the system
- **API**: Application Programming Interface - mechanisms allowing software systems to communicate
- **KPI**: Key Performance Indicator — metrics used to measure platform and business performance
- **CI/CD**: Continuous Integration / Continuous Deployment — automated software build and release pipeline
- **RBAC**: Role-Based Access Control — access restriction based on user roles
- **SLA**: Service Level Agreement — agreed performance and service standards
- **ISO**: International Organization for Standardization — a body defining international standards
- **FR**: Functional Requirement — describes what the system shall do
- **NFR**: Non-Functional Requirement — describes quality attributes and constraints
- **PR**: Performance Requirement — defines system performance expectations
- **DC**: Design Constraint — limitation affecting system design or implementation
- **RTM**: Requirements Traceability Matrix — maps requirements to design, testing, and validation artifacts
- **UML**: Unified Modeling Language — a standard for visualizing system design
- **QA**: Quality Assurance — processes ensuring software quality
- **SME**: Small and Medium Enterprise — typical provider category on the platform
- **CRUD API**: An API supporting create, read, update, and delete operations
- **Vendor/Provider**: A registered seller offering customized products
- **HTTPS/TLS**: Hypertext Transfer Protocol Secure / Transport Layer Security — a secure encrypted communication protocol

1.6 References

- ISO/IEC/IEEE 29148:2018
- ISO/IEC 25010:2011 (Software Quality Models)

2. General Description

2.1 Product Perspective

Customatch is a cloud-hosted, web-based system accessed through modern web browsers. It integrates user interfaces, business logic, data storage, analytics engines, and third-party services such as payment and logistics providers.

2.2 Product Functions

The system shall provide the following high-level functions:

- Guided visual product customization
- Provider discovery and matching
- Order creation, confirmation, and tracking
- Secure payments and transaction records
- Provider analytics dashboards
- Administrative monitoring and moderation

2.3 User Classes and Characteristics

2.3.1 Customers

- Non-technical end users
- Create and save designs
- View live visual
- View price estimates
- Chat with providers
- Place and track orders

2.3.2 Provider (Baker)

- Small to medium enterprises
- Manage profile and capabilities
- Define pricing rules
- Receive and respond to design requests
- Accept or reject orders
- View analytics

2.3.3 Administrators

- Manage users and providers
- Approve or suspend providers
- Monitor orders and disputes
- Access platform-wide analytics
- Responsible for platform configuration and compliance

2.4 Operating Environment

- Web browsers (Chrome, Firefox, Safari, Edge)
- Mobile and desktop devices
- Cloud-based servers and databases

2.5 Assumptions and Dependencies

The following assumptions and dependencies are considered during the definition of requirements for the Customatch platform:

- Users (customers and providers) have access to smartphones, tablets, or computers with modern web browsers.
- Users possess basic digital literacy sufficient to navigate web-based applications.
- Reliable internet connectivity is available for accessing core platform features.
- Third-party payment services (e.g., mobile money and digital wallets) are operational and accessible to users.
- Delivery and logistics services are available and capable of fulfilling orders within reasonable timeframes.

- Providers (bakers) are responsible for production quality, pricing accuracy, and timely fulfillment of orders.
- Providers maintain up-to-date availability, pricing, and customization capabilities on the platform.
- The platform does not directly handle food preparation, storage, or delivery logistics.
- Regulatory requirements related to food safety, consumer protection, and online transactions are enforced externally by relevant authorities.
- Initial user adoption will be limited during the MVP phase, allowing iterative improvements based on feedback.
- Platform analytics accuracy depends on consistent user and provider interaction with the system.
- Future expansion into additional product categories will reuse existing system architecture with minimal changes.

3. Functional Requirements

3.1 User Account Management

FR-1: The system shall allow users to register using email or phone number.

FR-2: The system shall authenticate users securely.

FR-3: Users shall be able to manage profile information.

3.2 Interactive Customization Studio (Core Module)

FR-4: The system shall allow users to specify all required cake customization attributes before placing an order.

- FR-4.1: The system shall allow users to select the cake shape.
- FR-4.2: The system shall allow users to specify cake size and number of tiers.
- FR-4.3: The system shall allow users to select one or more cake flavors.

- FR-4.4: The system shall allow users to select cake fillings.
- FR-4.5: The system shall allow users to choose icing type and icing color.
- FR-4.6: The system shall allow users to select decoration style and theme.
- FR-4.7: The system shall allow users to enter custom text inscriptions.
- FR-4.8: The system shall allow users to specify dietary preferences (e.g., eggless, sugar-free, allergen considerations).
- FR-4.9: The system shall allow users to specify the order quantity.
- FR-4.10: The system shall allow users to select a preferred delivery or pickup date.
- FR-4.11: The system shall allow users to upload reference images to guide customization.

FR-5: The system shall update the visual representation of the cake dynamically based on user selections.

FR-6: Users shall be able to modify customization selections prior to order submission.

- FR-6.1: The system shall enforce feasibility and validation rules to prevent unsupported or conflicting customization combinations based on provider capabilities.
- FR-6.2: The system shall provide clear feedback to users when invalid or unsupported combinations are selected.

3.3 Visualization Engine

FR-7: The system shall render a 2D visual preview for the cake

FR-8: Visuals shall reflect all selected attributes

FR-9: The rendering logic shall be component-based to support future 3D upgrades

3.4 Design Management

FR-10: Users shall be able to save customized cake designs for future use.

FR-11: Users shall be able to edit previously saved designs.

FR-12: Users shall be able to duplicate existing designs to create variations.

3.5 Provider Matching

FR-13: The system shall match user customization requests with eligible providers based on capabilities, availability, lead time, location, and pricing constraints.

FR-14: The system shall automatically exclude providers that cannot fulfill required customization attributes, delivery timelines, or feasibility rules.

FR-15: The system shall rank matched providers using configurable criteria, including price, ratings, delivery time, and relevance to the requested customization.

FR-16: The system shall present matched providers with key information, including estimated price, ratings, portfolios, and fulfillment timelines.

FR-17: The system shall allow users to filter, sort, and view detailed provider profiles before selecting a provider.

3.6 Order Management

FR-18: The system shall allow customers to place an order only after a provider has confirmed the availability and feasibility of the customized design.

FR-19: The system shall allow providers to respond to orders by accepting, rejecting, or requesting clarification from the customer.

FR-20: The system shall manage the order lifecycle using standardized statuses, including Pending, Accepted, In Progress, Completed, and Cancelled.

FR-21: The system shall automatically update and display order status changes in real time to both customers and providers based on workflow actions.

3.7 Pricing & Quotation Engine

FR-22: Calculate pricing dynamically based on selected attributes, provider pricing rules, and add-ons or complexity.

FR-23: Display minimum price, maximum price (if applicable), and final estimated price before order placement.

FR-24: Pricing logic configurable per provider.

3.8 Communication (Chat System)

FR-25: Allow direct messaging between customers and providers.

FR-26: Link chats to specific designs or orders.

FR-27: Persist message history.

3.9 Analytics and Insights

FR-28: Providers (MVP) shall access analytics for:

- Order volume
- Popular designs
- Conversion rate

FR-2: Administrators shall access platform analytics, including:

- Platform usage
- Provider performance
- Order success rates

4. Interface Requirements

4.1 User Interface

- The system shall provide a guided, step-by-step customization process that walks users through each attribute selection, providing real-time visual feedback and contextual tips.
- The interface shall be responsive and mobile-friendly, adapting layouts for smartphones, tablets, and desktop screens without loss of functionality or clarity.
- Error messages and prompts shall be clear, concise, and provide guidance on corrective actions.

4.2 Software Interfaces

- The system shall integrate with payment gateways (e.g., mobile money, card payments) for secure transaction processing.
- The system shall integrate with logistics and delivery service APIs to capture shipping and fulfillment details.
- External APIs for additional payment methods and logistics services shall be integrated in Phase 2.

4.3 Communication Interfaces

- All data exchanges between clients, servers, and third-party services shall use HTTPS/TLS for encryption and secure transmission.
- The system shall ensure all authentication tokens, session data, and sensitive information are transmitted securely.

5. Performance Requirements

PR-1: The system shall support concurrent users without noticeable degradation.

PR-2: Customization changes shall be reflected within acceptable response times.

PR-3: Order transactions shall be processed reliably.

PR-4: Page load time shall be ≤ 3 seconds.

PR-5: Pricing calculations shall be completed in ≤ 1 second.

PR-6: The system shall support ≥ 5,000 concurrent users for the MVP phase.

PR-7: Chat messages between users and providers shall be delivered in near real-time.

6. Security Requirements

- All data transmitted between clients and servers shall be encrypted using HTTPS/TLS.
- The system shall enforce secure authentication and authorization, including role-based access control (RBAC) for administrative features.
- All user inputs shall be validated on both the client and server sides to prevent injection attacks and other malicious activity.
- The system shall implement measures to protect against unauthorized data access, ensuring that users and providers can only access data they are permitted to view.

7. Design Constraints

DC-1: The MVP scope is limited to cake customization.

DC-2: The system architecture shall be modular to allow future expansion into additional product categories such as furniture, fashion, and event design.

DC-3: All components shall comply with applicable e-commerce, data protection, and consumer protection regulations.

DC-4: The system shall be compatible with major web browsers (Chrome, Firefox, Safari, Edge) and mobile devices.

DC-5: Initial development resources, including budget and personnel, are limited, requiring prioritization of core features.

DC-6: Third-party integrations for payment and logistics services shall adhere to their respective APIs and security requirements.

DC-7: The system shall enforce feasibility and validation rules to prevent unsupported customization combinations based on provider capabilities.

DC-8: UI/UX design shall follow accessibility standards to support diverse user groups.

DC-9: Data storage and retrieval operations shall comply with cloud service limitations and performance expectations.

DC-10: The architecture must support multi-category expansion beyond cakes.

DC-11: The codebase shall be modular and maintainable to facilitate updates and enhancements.

DC-12: Open-source tooling and frameworks are preferred for system development and maintenance.

8. Non-Functional Attributes

8.1 Usability

- The system shall require minimal learning effort for first-time users.
- Clear navigation, prompts, and validation feedback shall enhance user experience.

8.2 Reliability and Availability

- The system shall maintain data consistency and recover gracefully from failures.
- Uptime shall meet the agreed SLA for MVP (e.g., ≥ 99.5%).

8.3 Security

- All user data shall be encrypted in transit using HTTPS/TLS.
- Secure authentication mechanisms, including strong passwords and optional 2FA, shall be enforced.
- Role-Based Access Control (RBAC) shall limit administrative actions to authorized users.

8.4 Scalability

- The architecture shall support expansion into additional customization categories.
- The system shall handle increasing numbers of users, providers, and orders with minimal changes.

8.5 Maintainability

- The system shall support modular updates and enhancements.
- Code and system architecture shall follow best practices to reduce technical debt.

9. Appendices

Appendix A: Reference Materials

- **ISO/IEC/IEEE 29148:2018** – Systems and Software Engineering — Life Cycle Processes — Requirements Engineering
- **ISO/IEC 25010:2011** – Software Quality Models
- **Payment Gateway Documentation** – APIs for mobile money integrations
- **Logistics Service Documentation** – APIs and protocols for delivery services
- **UI/UX Design Mockups** – Clickable prototypes and wireframes
- **Platform Analytics Reports** – Examples of provider dashboards, order trends, and performance metrics

Appendix B: Acronyms and Abbreviations (Supplemental)

- **SRS**: Software Requirements Specification
- **MVP**: Minimum Viable Product
- **PWA**: Progressive Web Application
- **UAT**: User Acceptance Testing
- **RBAC**: Role-Based Access Control
- **CRUD**: Create, Read, Update, Delete
- **HTTPS/TLS**: Secure communication protocol
- **FR**: Functional Requirement
- **NFR**: Non-Functional Requirement
- **PR**: Performance Requirement
- **DC**: Design Constraint

10. Uses of the SRS Document

The SRS serves as a **single source of truth** for all stakeholders and provides value in multiple areas:

- **Development and Implementation:** Guides software engineers and designers in building the Customatch platform according to agreed functional and non-functional requirements.

- **Testing and Quality Assurance:** Supports QA teams in creating test plans, conducting UAT, and ensuring each requirement is verifiable and traceable.

- **Project Planning and Management:** Enables project managers to plan schedules, allocate resources, and monitor progress against defined requirements.

- **Stakeholder Communication:** Communicates system capabilities, workflows, and expectations to business stakeholders, investors, and partners.

- **Regulatory and Compliance Review:** Provides a structured reference for auditors, compliance officers, and legal teams to verify adherence to relevant e-commerce, data protection, and consumer protection standards.

11. Conclusion

This Software Requirements Specification provides a comprehensive, detailed, and testable framework for the development of the Customatch platform. It defines the functional and non-functional requirements, interface specifications, performance standards, design constraints, and security measures needed to deliver a secure, reliable, and scalable system.

By following this SRS, all stakeholders—developers, designers, testers, project managers, business leaders, and investors—can ensure alignment between the platform's technical implementation and business objectives, facilitating successful delivery of the MVP for cake customization and laying the groundwork for future expansion into multiple product categories.

The document supports traceability, verifiability, and scalability, enabling iterative improvements, data-driven decision-making, and an enhanced user experience while reducing project risks and ensuring consistent quality.